

fn match_datetime_component

Michael Shoemate (@Shoeboxam)

This proof resides in “contrib” because it has not completed the vetting process.

Proves soundness of `match_datetime_component` in `mod.rs` at commit `f5bb719` (outdated¹).

`match_datetime_component` returns the data type and number of possible unique values of a compute operation that retrieves a datetime component.

1 Hoare Triple

Precondition

Compiler-verified

- Argument `temporal_function` of type `TemporalFunction`

Caller-verified

None

Pseudocode

```
1 def match_datetime_component (
2     temporal_function: TemporalFunction,
3 ) -> DataType | None:
4     return (
5         {
6             Millennium: DataType.UInt32,
7             Century: DataType.Int32,
8             Year: DataType.Int32,
9             IsoYear: DataType.Int32,
10            Quarter: DataType.Int8,
11            Month: DataType.Int8,
12            Week: DataType.Int8,
13            WeekDay: DataType.Int8,
14            Day: DataType.Int8,
15            OrdinalDay: DataType.Int16,
16            Hour: DataType.Int8,
17            Minute: DataType.Int8,
18            Second: DataType.Int8,
19            Millisecond: DataType.Int32,
20            Microsecond: DataType.Int32,
21            Nanosecond: DataType.Int32,
22        }
23    ).get(temporal_function)
```

¹See new changes with `git diff f5bb719..adf030e rust/src/transformations/make_stable_expr/namespace_dt/expr_datetime_component/`
`rs`

Postcondition

Theorem 1.1. For any choice of `temporal_function`, returns the data type of the output if the input is an infallible row-by-row temporal expression, otherwise returns none.

Proof. For each choice of `temporal_function`, the data type is directly verified via testing. Infallible row-by-row is verified by inspection of the Polars source code. $\square$